Electricity numericals2

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Medium level Difficulty

Part 3

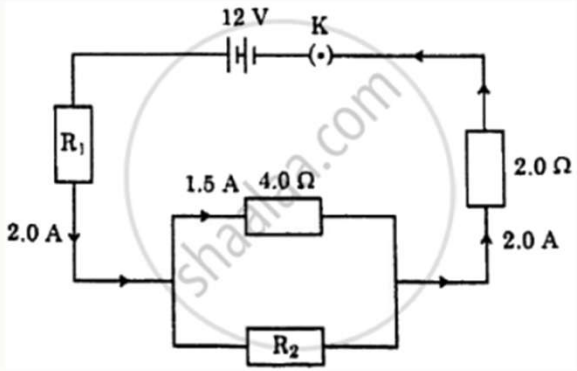
Question1: Several electric bulbs designed to be used on a 220 Velectric supply line are rated 10 W. How many lamps can be connected in parallel with each other across the two wires of the 220 V line if the maximum allowable current is 5 A?

Question2: In a house, 2 bulbs of 50 W each are used for 6 hours daily, and an electric geyser of 1 kW is used for 1 hour daily. Calculate the total energy consumed in a month of 30 days and its cost at the rate of ₹8.00per kWh.

Question 3: Calculate the length of a wire of radius 0.01 cm, resistance 7 Ω and resistivity 44 × 10^ {-6} Ω · m.

Question 4: A copper wire has a diameter of 0.5 mm and a resistivity of $1.6 \times 10^{-8}\ \Omega$m. What will be the length of this wire to make its resistance $10\ \Omega$? How much does the resistance change if the diameter is doubled?

Question 5:



i. The given electric circuit is a part of an electrical device. Use the information given in the electric circuit diagram to calculate:

I. Potential difference across the ends of resistor R_2.

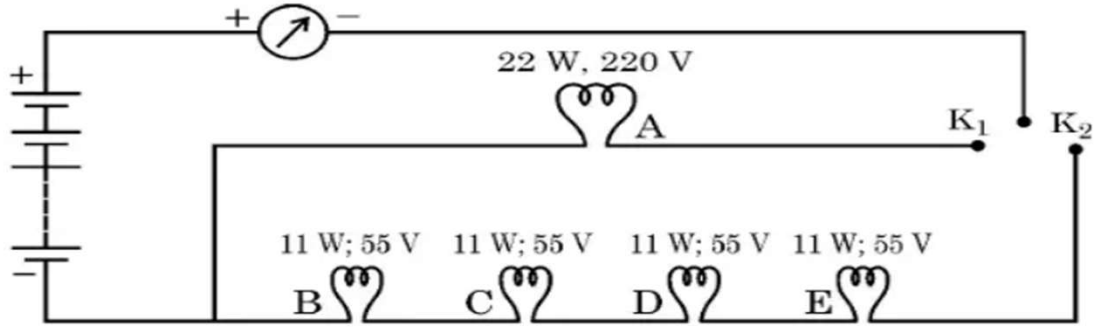
II. Value of resistor R_1.

Difficult or Lengthy Questions

Part 4

Question 6

In a domestic circuit five LED bulbs are arranged as shown. The source voltage is 220 V and the power rating of each bulb is marked in the circuit diagram. Based on the following circuit diagram, answer the following questions:



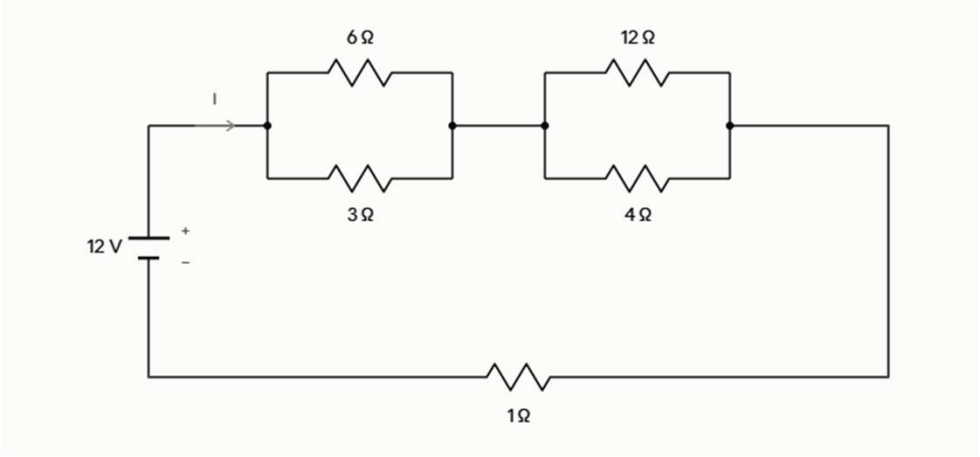
(a) State what happens when
(i) key K_1 is closed.
(ii) key K_2 is closed.
(b) Find the current drawn by the bulb B when it glows.
(c) Calculate
(i) the resistance of bulb B, and
(ii) total resistance of the combination of four bulbs B, C, D and E.
OR
(c) What would happen to the glow of all the bulbs in the circuit when keys K_1 and K_2 both are closed and the bulb C suddenly get fused ? Give reason to justify your answer.

Question 7: An electric oven is designed to work on the mains voltage of 220 V. This oven consumes 11 units of electrical energy in 5 hours.

Answer the following:

1. Define electric power. Express it in terms of potential difference V and resistance R.
2. Calculate the power rating of the oven.
3. Calculate the current drawn by the oven.
4. Calculate the resistance of the oven when it is red hot.

Question8: A 12 V battery (negligible internal resistance) is connected to this network: 6 Ω and 3 Ω in parallel, in series with 12 Ω and 4 Ω in parallel, in series with a 1 Ω resistor. Find (a) the total resistance, (b) the current through each of the five resistors, and (c) the power dissipated in the 1 Ω resistor.



Question9: A house on a 220 V supply has 5 bulbs of 40 W (used 6 h/day), 3 fans of 75 W (10 h/day), a 2 kW geyser (1 h/day) and a 400 W cooler (8 h/day).
(a) Find the energy used in 30 days.
(b) Find the bill at ₹6 per unit.
(c) If everything runs at once, what current flows, and which fuse (5 A, 10 A, 15 A, 30 A) is suitable?

Question10: A 1.1 kW–220 V heater has a nichrome coil of resistivity 1.0×10^{-6} Ω m and diameter 0.4 mm.

(a) Find the coil's resistance and length.

(b) With 80% efficiency, how long does it take to heat 2 kg of water from 20 °C to 70 °C? (amount of that required to raise temperature of 1Kg water by 1 °C or 1k is Specific heat of water = 4200 J kg^{-1} K^{-1}.)

(a) (c) What power does it deliver if the supply drops to 110 V?

Q1) Find the current drawn by a single bulb (I_1):

- Using the formula $P = V \times I$:
- $I_1 = \frac{P}{V} = \frac{10\ \text{W}}{220\ \text{V}} = \frac{1}{22}\ \text{A}$
- **Calculate the number of bulbs (n):**
- If $n$bulbs are connected in parallel, the total current (I) is the sum of the individual currents:
- $I = n \times I_1$
- $5\ \text{A} = n \times \frac{1}{22}$
- $n = 5 \times 22 = 110$

Q2) Calculate daily energy consumption (E_{daily}):

- Energy by bulbs: 2×50 W $\times 6$ h $= 600$ Wh $= 0.6$ kWh
- Energy by geyser (1 kW $= 1000$ W): 1000 W $\times 1$ h $= 1000$ Wh $= 1.0$ kWh
- Total daily energy = $0.6 + 1.0 = 1.6$ kWh

- **Calculate monthly energy consumption (E_{monthly}):**
- $E_{\text{monthly}} = 1.6$ kWh/day $\times 30$ days $= 48$ kWh
- **Calculate total cost:**
- Cost $= 48$ kWh $\times$ ₹8.00/kWh $=$ ₹384.00

Q3) Given: $r = 0.01\ cm = 10^{-4}\ m$, $R = 7\ \Omega$, $\rho = 44 \times 10^{-6}\ \Omega\ m$

- **Formula:** $R = \rho l/A$, so $l = RA/\rho$
- **Area of cross-section** (taking $\pi = 22/7$):
- $A = \pi r^2 = (22/7) \times (10^{-4})^2 = (22/7) \times 10^{-8}\ m^2$
- **Length:**
- $l = (7 \times (22/7) \times 10^{-8}) / (44 \times 10^{-6})$
- $l = (22 \times 10^{-8}) / (44 \times 10^{-6})$
- $l = 0.5 \times 10^{-2}\ m$
- **$l = 5 \times 10^{-3}\ m = 5\ mm$**

Q4) Find the cross-sectional area (A):

- Radius (r) = $\frac{d}{2} = \frac{0.5 \text{ mm}}{2} = 0.25 \text{ mm} = 2.5 \times 10^{-4}$ m
- $A = \pi r^2 = \frac{22}{7} \times (2.5 \times 10^{-4})^2 \approx 1.963 \times 10^{-7}$ m^2

- **Calculate length (l):**
- Using the formula $R = \rho \frac{l}{A}$:
- $l = \frac{R \times A}{\rho} = \frac{10 \times 1.963 \times 10^{-7}}{1.6 \times 10^{-8}} \approx 122.7$ m
- **Effect of doubling the diameter:**
- Since $R \propto \frac{1}{A}$ and $A \propto d^2$, we get $R \propto \frac{1}{d^2}$.
- If the diameter is doubled ($d' = 2d$), the area increases by 4 times, making the new resistance **one-fourth** of the original resistance ($R' = \frac{10}{4} = 2.5\ \Omega$).
- **Answer:** Length = 122.7 m; New resistance drops to $2.5\ \Omega$.

Q5) I. Potential difference across the ends of resistor R_2

- The potential difference across the parallel combination equals the voltage across the $4.0\ \Omega$resistor.
- $V = I \times R = 1.5\ \text{A} \times 4.0\ \Omega = \mathbf{6.0}\ \text{V}$.
- **II. Value of resistor R_1**
- **Total resistance of the circuit:**
- $R_{\text{total}} = \frac{V_{\text{total}}}{I_{\text{total}}} = \frac{12\ \text{V}}{2.0\ \text{A}} = 6.0\ \Omega$
- **Resistance of the parallel branch:**
- $R_p = \frac{V_p}{I_{\text{total}}} = \frac{6.0\ \text{V}}{2.0\ \text{A}} = 3.0\ \Omega$
- **Finding R_1(series connection):**
- $R_{\text{total}} = R_1 + R_p + 2.0\ \Omega$
- $6.0\ \Omega = R_1 + 3.0\ \Omega + 2.0\ \Omega$
- $R_1 = \mathbf{1.0\ \Omega}$

Q6(a)

- **(i) Key K_1 closed:** Bulb A glows.
- **(ii) Key K_2 closed:** Bulbs B, C, D, and E glow.
- *(b)*
- **Current drawn by bulb B:** 0.2 A(calculated as $I = \frac{P}{V} = \frac{11\ \text{W}}{55\ \text{V}}$).
- *(c)*
- **(i) Resistance of bulb B:** $275\ \Omega$(calculated as $R = \frac{V^2}{P} = \frac{55^2}{11}$).
- **(ii) Total resistance of B, C, D, and E:** $1100\ \Omega$(four identical series resistors: $4 \times 275\ \Omega$).
- **OR (c)**
- **Glow of bulb A:** Remains unaffected and glows normally.
- **Glow of bulbs B, D, and E:** They stop glowing because the fusing of bulb C breaks the series circuit path in that branch, stopping the current flow.

Q7.Given data: Energy consumed (E) = 11 units $=$ 11 kWh Time (t) = 5 hours

- **Formula:** Power $(P) = \frac{\text{Energy } (E)}{\text{Time } (t)}$
- **Calculation:** $P = \frac{11 \text{ kWh}}{5 \text{ h}} = 2.2 \text{ kW} = 2200 \text{ W}$
- **3. Calculate the current drawn by the oven.**
- **Given data:** Power (P) = 2200 W Voltage (V) = 220 V
- **Formula:** $P = V \times I \Longrightarrow I = \frac{P}{V}$
- **Calculation:** $I = \frac{2200 \text{ W}}{220 \text{ V}} = 10 \text{ A}$
- **4. Calculate the resistance of the oven when it is red hot.**
- **Given data:** Voltage (V) = 220 V Current (I) = 10 A
- **Formula (Ohm's Law):** $R = \frac{V}{I}$
- **Calculation:** $R = \frac{220 \text{ V}}{10 \text{ A}} = 22 \ \Omega$
- *(Alternatively, using* $P = \frac{V^2}{R} \Longrightarrow R = \frac{V^2}{P} = \frac{220^2}{2200} = 22 \ \Omega$*)*

- **Q8.**
- (a) The first parallel group is 2 Ω and the second is 3 Ω, so R = 2 + 3 + 1 = **6 Ω**.
- (b) I = 12/6 = **2 A** in the 1 Ω resistor. The first group has 4 V across it, so 6 Ω carries **2/3 A** and 3 Ω carries **4/3 A**. The second group has 6 V across it, so 12 Ω carries **0.5 A** and 4 Ω carries **1.5 A**.
- (c) $P = I^2R = 4 \times 1 =$ **4 W**.

- **Q9.**
- (a) Daily energy = 1.2 + 2.25 + 2 + 3.2 = 8.65 kWh. Over 30 days that is **259.5 units**.
- (b) Bill = 259.5 × 6 = **₹1557**.
- (c) Total power = 200 + 225 + 2000 + 400 = 2825 W, so I = 2825/220 ≈ **12.8 A**. A **15 A fuse** is suitable.

- **Q10.**
- (a) $R = V^2/P = 220^2/1100 =$ **44 Ω**. The area is $A = \pi d^2/4 \approx 1.26 \times 10^{-7}\ m^2$, so $l = RA/\rho \approx$ **5.5 m**.
- (b) The heat needed is $Q = 2 \times 4200 \times 50 = 4.2 \times 10^5$ J. Useful power is $0.8 \times 1100 = 880$ W, so $t = 4.2 \times 10^5/880 \approx$ **477 s (about 8 min)**.
- (c) $P = 110^2/44 =$ **275 W**. Halving the voltage cuts the power to one-fourth.